

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Trigonometric Equations

Trig equations are interval questions: transform first, then find every valid angle.

TRIGONOMETRIC EQUATIONS · P2 · 10

ELITEIGCSE.COM · WHATSAPP +20 112 000 9622



DR ESLAM AHMED · CAIRO UNIVERSITY FACULTY OF ENGINEERING

The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA12 website answer bank. The topic reduces to these exam moves, with hard variants kept visible on each page.

01 CAST: Quadrant Solutions

TRIGGER *solve trig equation*

02 IDENT: Identity First

TRIGGER *quadratic trig equation or mixed functions*

03 PERIOD: Use Period

TRIGGER *solutions beyond one cycle*

04 TRANS: Transformed Trig

TRIGGER $a \sin(bx + c) + d = 0$

BANK EVIDENCE Local pages: 19 problem, 38 answer, 21 notes. Website primary entries: 20.

01 CAST: Quadrant Solutions

TRIGGER *solve trig equation*

BECOMES Find reference angle and all quadrants.

FIRST LINE TO WRITE

$$\sin x = a, \quad \cos x = a, \quad \tan x = a$$

SIMPLEST STRATEGY

- 1 Calculate reference angle.
- 2 Ask sign.
- 3 Select quadrants.
- 4 Turn into interval solutions.

WORKED MODEL

$$\cos x = -\frac{1}{2} \Rightarrow x = 120^\circ, 240^\circ.$$

HARD VARIANTS

- 1 For kx , solve over the expanded interval for kx , then divide.
- 2 Use radians if the interval is in radians.
- 3 Remove solutions outside the requested interval.

— BOTTOM LINE

One calculator angle is rarely the whole answer.

02 IDENT: Identity First

TRIGGER *quadratic trig equation or mixed functions*

BECOMES Use identities to make one trig function.

FIRST LINE TO WRITE

$$\sin^2 x + \cos^2 x = 1$$

SIMPLEST STRATEGY

- 1 Identify mixed functions.
- 2 Decide identity.
- 3 Exchange to one function.
- 4 Normal factorise or solve.
- 5 Test interval.

WORKED MODEL

$$2 \sin^2 x + \sin x - 1 = 0.$$

HARD VARIANTS

- 1 If factorising, set each factor equal to zero.
- 2 Do not divide by $\sin x$ or $\cos x$ unless you handle zero separately.
- 3 Check every root against the original interval.

— BOTTOM LINE

A trig quadratic is still a quadratic after substitution.

03 PERIOD: Use Period

TRIGGER *solutions beyond one cycle*

BECOMES Generate repeated solutions by adding periods.

FIRST LINE TO WRITE

$$x = \alpha + 360^\circ k$$

SIMPLEST STRATEGY

- 1 Pick base solutions.
- 2 Establish period.
- 3 Repeat using k .
- 4 Interval filter.
- 5 Order answers.
- 6 Degree/radian units.

WORKED MODEL

| For $\tan x = 1$, $x = 45^\circ + 180^\circ k$.

HARD VARIANTS

- 1 Tangent period is 180° or π .
- 2 Transformed functions have period divided by the inside multiplier.
- 3 List solutions in increasing order.

— BOTTOM LINE

Period gives the missing angles.

SAME IDEA Use PERIOD: Use Period.

QUESTION

| Solve $\tan x = \sqrt{3}, 0 \leq x \leq 720^\circ$.

YOUR SOLUTION

04 TRANS: Transformed Trig

TRIGGER $a \sin(bx + c) + d = 0$

BECOMES Isolate the trig part, then solve the inside angle.

FIRST LINE TO WRITE

$$bx + c = \theta$$

SIMPLEST STRATEGY

- 1 Tidy to one trig expression.
- 2 Read the inside angle.
- 3 Allow expanded interval.
- 4 Normal CAST solve.
- 5 Scale back to x .

WORKED MODEL

$$\sin(2x) = \frac{1}{2} \Rightarrow 2x = 30^\circ, 150^\circ, \dots$$

HARD VARIANTS

- 1 Expand the interval for the inside expression before finding angles.
- 2 After dividing back, check endpoints carefully.
- 3 Vertical shifts must be removed before CAST.

— BOTTOM LINE

Solve the inside angle first.

Practice 04

SAME IDEA Use TRANS: Transformed Trig.

QUESTION

| Solve $\sin(2x - 30^\circ) = 0.5$ for $0 \leq x \leq 180^\circ$.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
solve trig equation	$\sin x = a, \quad \cos x = a, \quad \tan x = a$
quadratic trig equation or mixed functions	$\sin^2 x + \cos^2 x = 1$
solutions beyond one cycle	$x = \alpha + 360^\circ k$
$a \sin(bx + c) + d = 0$	$bx + c = \theta$